

NYERI, Kenya

WMO: 637170

Lat: 0.4421S

Lon: 36.9808E

Elev: 1798

StdP: 81.51

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	10.8	5.4	6.9	19.2	7.1	7.8	18.4	7.4	18.8	6.2	19.1	1.3	230	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	12.2	27.5	16.2	26.6	16.4	25.7	16.5	19.0	23.4	18.5	23.0	18.2	22.7	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.7	20.8	17.1	15.3	20.2	16.8	15.0	19.8	62.4	23.5	60.8	23.1	59.5	22.8	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.0	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			6.9	20.2	1.4	0.8	5.8	20.7	5.0	21.2	4.2	21.6	3.2	22.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	18.5	19.3	19.6	19.6	18.8	17.6	16.5	16.7	18.2	19.2	18.4	18.2
	DBStd	1.57	1.14	1.28	1.39	1.12	0.78	1.17	1.40	1.44	1.44	1.07	1.06	1.11
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	219	12	4	4	2	4	28	59	52	19	4	12	16
	CDD10.0	3054	262	260	298	289	272	227	201	208	247	286	252	254
	CDD18.3	233	16	32	44	41	19	5	2	2	16	32	13	12
	CDH23.3	1144	179	314	317	102	9	3	3	4	61	71	24	58
	CDH26.7	78	4	32	30	8	0	0	0	1	1	1	1	1
(14) Wind	WSAvg	2.2	2.1	2.1	2.3	2.2	2.2	2.1	2.0	2.3	2.7	2.4	2.1	2.1
(15) Precipitation	PrecAvg	1129	80	60	88	187	167	45	42	56	39	129	151	88
	PrecMax	1497	248	145	242	312	340	83	65	115	80	327	302	144
	PrecMin	794	35	25	23	109	82	14	22	27	21	23	88	29
	PrecStd	182	55	31	48	64	75	16	14	20	16	76	60	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	29.0	28.8	27.1	24.2	23.9	23.7	23.8	26.4	26.3	25.1	26.0
		MCWB	15.5	15.2	16.1	17.6	18.2	17.3	16.5	16.0	16.2	16.4	17.3	16.2
	2%	DB	26.2	27.6	27.5	25.4	23.5	22.5	22.2	22.5	25.0	25.0	24.0	24.8
		MCWB	16.0	15.8	16.6	17.5	18.0	17.1	16.0	16.0	16.1	16.4	17.1	16.7
	5%	DB	25.3	26.5	26.4	24.4	22.8	21.6	21.1	21.4	23.9	24.0	23.1	23.8
		MCWB	16.3	16.2	16.6	17.5	17.7	16.8	15.8	15.7	16.0	16.5	17.0	16.7
	10%	DB	24.3	25.4	25.3	23.4	22.0	20.8	19.9	20.2	22.8	23.1	22.2	22.8
		MCWB	16.5	16.4	16.7	17.5	17.5	16.4	15.4	15.4	15.8	16.3	16.9	16.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.2	19.2	19.5	19.1	18.5	17.7	17.6	17.9	18.4	18.6	19.1
		MCDB	23.8	23.6	24.7	23.9	22.9	22.3	22.0	22.1	23.8	23.2	22.7	22.9
	2%	WB	18.2	18.5	18.5	18.7	18.5	17.7	16.8	16.8	17.1	17.7	18.0	18.2
		MCDB	23.1	23.6	24.0	23.0	22.3	21.3	20.8	21.0	22.8	22.3	21.9	22.1
	5%	WB	17.7	17.9	18.1	18.2	18.1	17.1	16.2	16.2	16.6	17.3	17.6	17.8
		MCDB	22.5	23.6	23.5	22.5	21.9	20.6	19.9	20.2	22.0	21.7	21.3	21.6
	10%	WB	17.2	17.3	17.6	17.9	17.7	16.7	15.6	15.6	16.1	17.0	17.2	17.3
		MCDB	22.0	22.9	22.8	22.0	21.2	19.8	18.9	19.2	21.3	21.2	20.8	21.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	13.8	12.2	8.6	7.2	6.9	6.9	7.4	9.2	8.8	8.4	10.5
		MCDBR	14.7	15.9	14.6	10.1	8.1	8.2	9.2	9.4	11.2	10.2	10.0	12.3
		MCWBR	6.5	6.5	5.8	3.8	3.4	3.6	4.1	4.1	4.1	3.4	4.2	5.7
	5% WB	MCDBR	12.1	12.7	11.2	8.7	7.5	7.5	8.1	8.5	9.7	8.7	8.6	9.9
		MCWBR	6.1	5.8	5.0	3.7	3.4	3.7	4.1	4.0	4.1	3.4	4.0	5.1
(40) Clear-Sky Solar Irradiance	taub		0.356	0.378	0.373	0.331	0.329	0.388	0.405	0.378	0.350	0.352	0.347	0.353
	taud		2.470	2.388	2.433	2.610	2.605	2.387	2.313	2.393	2.469	2.482	2.522	2.498
	Ebn at Noon		974	953	949	969	947	876	866	911	956	967	975	972
	Edh at Noon		115	127	121	98	95	116	126	121	115	114	109	111
(44) All-Sky Solar Radiation	RadAvg		6.20	6.54	6.33	5.62	5.42	4.82	4.54	4.80	5.75	5.74	5.26	5.63
	RadStd		0.48	0.36	0.48	0.36	0.20	0.32	0.42	0.31	0.30	0.28	0.32	0.40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page